1 Question 1

ICP performance discussion:

- *bunny_original.ply* (reference) and *bunny_perturbed.ply* (aligned) (Fig. 1): ICP performs great on this pair and closely aligns the clouds with an RMS value of 1×10^{-8}. Since the perturbation is small and the clouds overlaps well, the algorithm converges to the correct solution.
- *bunny_original.ply* (reference) and *bunny_returned.ply* (aligned) (Fig. 2): The initial clouds do not intersect and have large differences in both position and orientation. As a result, ICP fails to recover the correct alignment in this setting. The reported RMS of 0.013 indicates that the algorithm has converged to a local minimum, which explains the incorrect alignment that we observed.
- *Notre_Dame_Des_Champs_1.ply* (reference) and *Notre_Dame_Des_Champs_2.ply* (aligned) (Fig. 3): ICP performs well and produces a visually correct alignment. However, the RMS value of 1.33 remains relatively high. This can be explained by the presence of points in the aligned cloud (e.g., one building wall or a road perpendicular to the boulevard) that do not have true correspondences in the reference cloud (as these points are missing in that cloud). Consequently, despite the visually pleasing alignment, the RMS remains much higher than it was in the first example.

In the ICP algorithm, the reference cloud represents the one that stays fixed in its original form and coordinate frame. In contrast, the aligned cloud is rotated and translated iteratively so that its points match the closest points of the reference.

In the *Notre Dame* example, *Notre_Dame_Des_Champs_1.ply* should be the reference cloud, as it represents a much larger scene and contains more points than the second one. The second cloud can be considered a subset of the first, as it covers only a portion of the overall scene. If the second cloud is chosen as the reference, a large number of points from the aligned (first) cloud would be forced to match missing points, which would result in incorrect correspondences. This behavior is confirmed by the results shown in Fig. 3 (b), where the algorithm completely fails. In this case, the RMS reaches a value of 13.19, which is almost ten times higher than in the reverse setting.

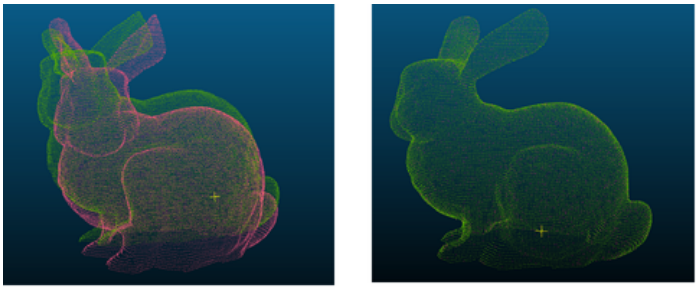


Figure 1: Results of CloudCompare ICP: (a) Initial clouds *bunny_original.ply* (green, reference) and *bunny_perturbed.ply* (red, aligned); (b) Clouds after registration.

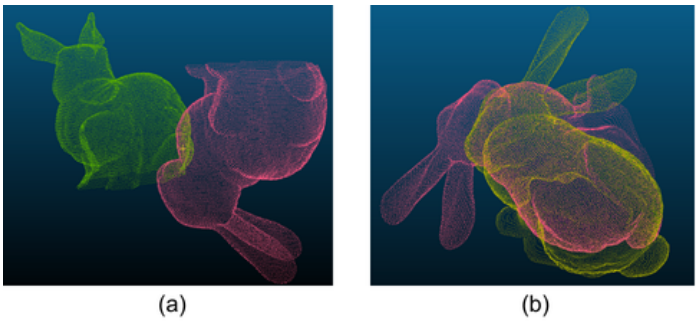


Figure 2: Results of CloudCompare ICP: (a) Initial clouds *bunny_original.ply* (green, reference) and *bunny_returned.ply* (red, aligned); (b) Clouds after registration.

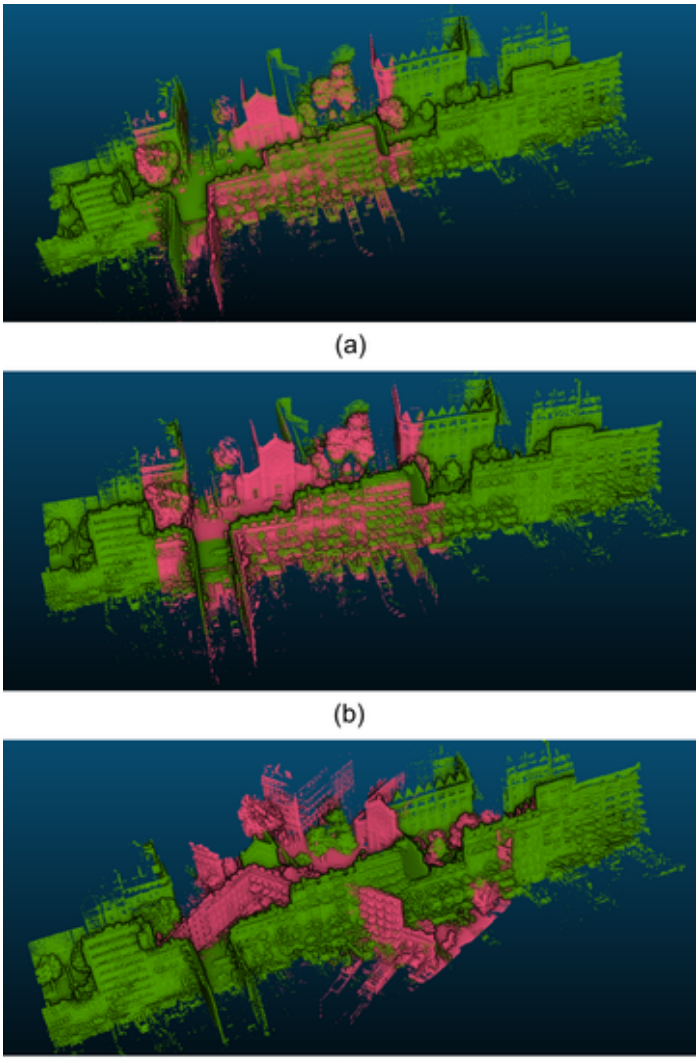


Figure 3: Results of CloudCompare ICP: (a) Initial clouds *Notre_Dame_Des_Champs_1.ply* (green) and *Notre_Dame_Des_Champs_2.ply* (red); (b) Clouds after registration when *Notre_Dame_Des_Champs_1.ply* is the reference cloud; (c) Clouds after registration when *Notre_Dame_Des_Champs_2.ply* is the reference cloud.

2 Question 2

Before the rigid transformation, the RMS error is **0.16**, and after applying the transformation, the error drops to $\mathbf{1.2 \times 10^{-8}}$. Based on these results, as well as the visualization in Fig. 4, we can conclude that the two clouds are perfectly aligned.

The alignment works in this setting because we can use the notion of sorted points to define the correspondences between the two clouds (i.e. $p_i \leftrightarrow p'_i$). This allows us to accurately minimize the distance (function) $\sum_{i=1}^{n} \|p_i - (R \times p'_i + T)\|^2$. On the other hand, CloudCompare ICP does not know or use these correspondences and must estimate them by computing nearest neighbor distances. As previously discussed, this estimation becomes inaccurate because of the different configurations of initial clouds, which therefore leads to suboptimal results.

The function would not work on *Notre Dame des Champs* clouds as it requires two clouds with the same number of (matching) points.

3 Question 3

Figure 5 shows the RMS errors during ICP convergence for: (a) *ref2D.ply* and *data2D.ply* and (b): *bunny_original.ply* and *bunny_perturbed.ply*. The maximum number of iterations in (a) and (b) is set to 10 and 25, respectively, and *RMS_threshold* parameter is set to 1×10^{-4} (*Note: The first iteration (at index 0) in the plots corresponds to*

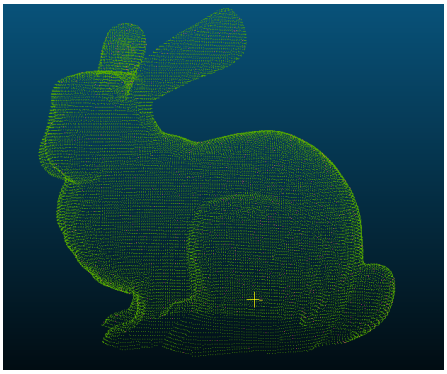


Figure 4: Results of best rigid transformation on *bunny_original.ply* and *bunny_returned.ply*.

the RMS error before any alignment is applied. To match the initial requirements, the number of iterations in the implementation was incremented by one).

4 Question 4

In both cases, we see the monotonic convergence of the ICP algorithm. Given our fixed number of matched points, these results follow the observations by [1]. More specifically:

- Case (a): The RMS error drops sharply to a value of 0.06 by the 6th iteration, after which the error is steady. Increasing the number of iterations does not further reduce the error, as it changes only on the 10-12th decimal level and never falls below the aforementioned threshold. The visual results shown in Fig.6 confirm this behavior, as the 6th, 7th, and 10th iterations look visually identical. Based on this, we can conclude that the given solution is stable and produces correct alignment.
- Case (b): The RMS error again decreases, but at a noticeably slower rate. However, the error eventually falls below the specified threshold at the 19th iteration, at which point the algorithm is stopped. This perfect alignment is shown in Fig. 7.

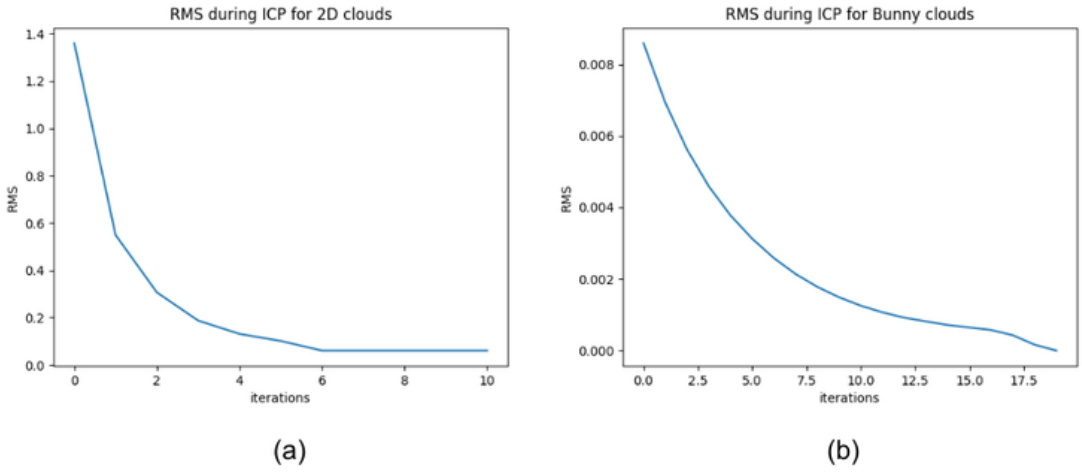


Figure 5: RMS during ICP convergence for: (a) *ref2D.ply* and *data2D.ply*; (b) *bunny_original.ply* and *bunny_perturbed.ply*.

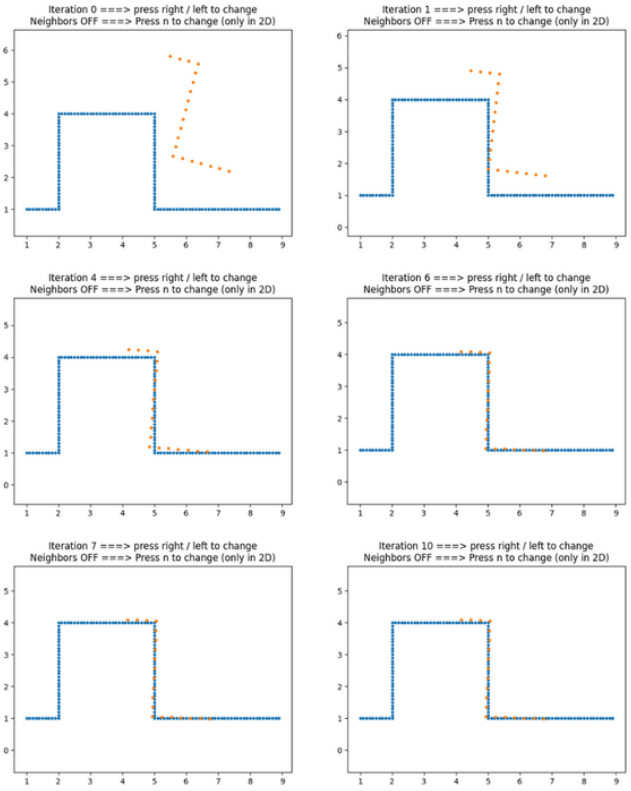


Figure 6: *ref2D.ply* and *data2D.ply* during ICP alignment through iterations.

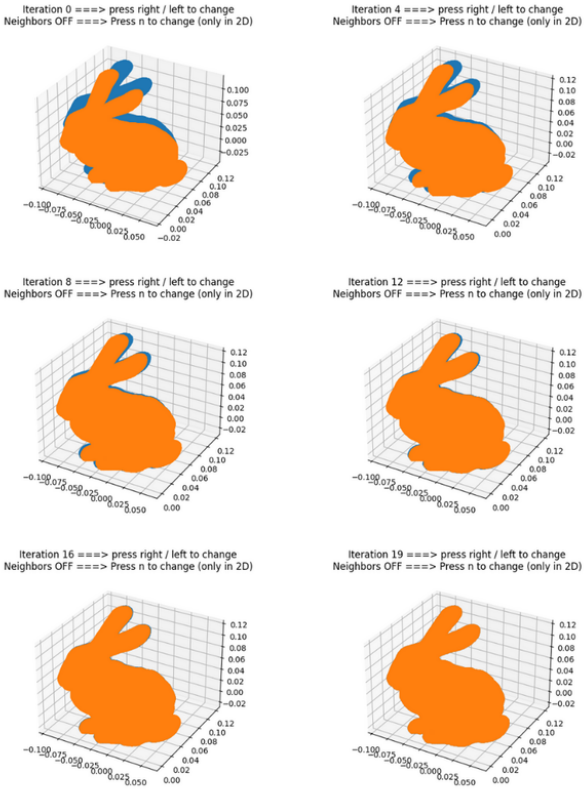


Figure 7: *bunny_original.ply* and *bunny_perturbed.ply* during ICP alignment through iterations.

5 Question 5

Figure 8 shows the RMS error during fast ICP convergence for *Notre Dame* clouds. The maximum number of iterations is set to 100, and *RMS_threshold* parameter is set to 1×10^{-4}.

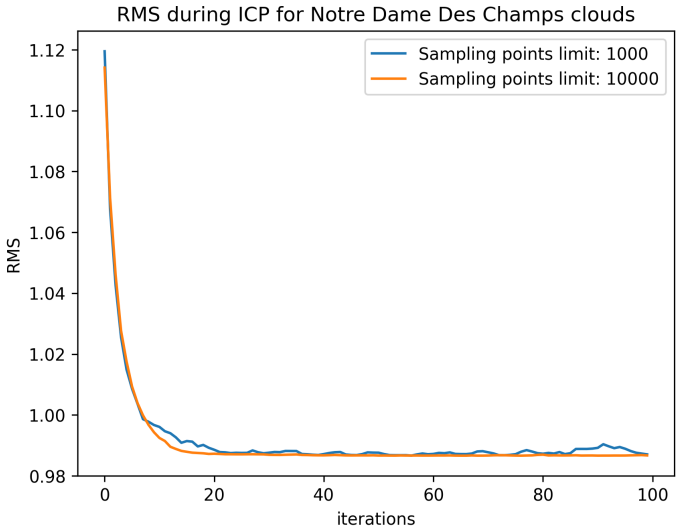


Figure 8: *Notre_Dame_Des_Champs_1.ply* and *Notre_Dame_Des_Champs_2.ply* during fast ICP alignment through iterations with search of: (a) 1000 neighbors (blue); (b) 10000 neighbors (orange).

In both cases, the RMS error drops sharply over the initial iterations (up to the 15th-18th iteration), after which it remains steady. Moreover, the error never drops below the threshold, even when the number of iterations is increased, which was expected and has already been discussed. Since the values are close to the ones returned by CloudCompare ICP in Question 1, we can consider the results to be correct.

When using a smaller number of neighbors, the error curve is noisier and shows slight variability. With 10000 neighbors, we observe a more stable curve. This behavior is expected, as using a larger number of neighbors provides a better approximation of the alignment over the entire point set, which results in more stable RMS error across all points. The mean, i.e., the RMS error, remains similar in both cases (around 0.99-1.00).

References

[1] Paul J. Besl and Neil D. McKay. A method for registration of 3-d shapes. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 14(2):239–256, February 1992.